# Xianhao Carton Liu

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## Education

**Computer Science Ph.D. Candidate** — ***University of Minnesota, Twin Cities***
*Advisor:* ***Zhu-Tian Chen*** — *Aug 2024 – Now*
Research topic: Human AI Interaction in Physical Environments and Real World

**B.Eng. with Honors in Artificial Intelligence (Turing Class)** — ***Zhejiang University***
*Outstanding honored undergraduate in* ***Chu Kochen Honors College*** — *Sep 2020 – June 2024*

## Publications

**Generative Proxy: Synthesizing Proxy-Based Interfaces for Real-World Interaction Across AR Glasses**
*Accepted in UIST 2026 [Paper] [Video]* — *Jun 2026*
- Xianhao Carton Liu, William Chastek, Eric J Gonzalez, Mar Gonzalez-Franco, Chen Zhu-Tian

**Can AR Embedded Visualizations Foster Appropriate Reliance on AI in Spatial Decision Making? A Comparative Study of AR X-Ray vs. 2D Minimap**
*Accepted in CHI 2026 [Paper] [Video] [Github]* — *Jan 2026*
- Xianhao Carton Liu, Difan Jia, Tongyu Nie, Evan Suma Rosenberg, Victoria Interrante, Chen Zhu-Tian

**Reality Proxy: Fluid Interactions with Real-World Objects in MR via Abstract Representations**
*Accepted in UIST 2025 [Paper] [Video]* — *May 2025*
- Xiaoan Liu, Difan Jia, Xianhao Carton Liu, Mar Gonzalez-Franco, Chen Zhu-Tian

**Understanding User Reliance on AI with Eye-Tracking Smart Glasses in Procedural Tasks**
*Under review in CHI 2027* — *Sep 2026*
- Xianhao Carton Liu, Bingkun Xie, Ruiheng Wang, Zhiyuan (Ryan) Wang, Chen Zhu-Tian

## Research Experience

**Harvard Visual Computing Group, Harvard University** — ***Cambridge, USA***
*Research intern advised by* ***Zhu-Tian Chen*** *and* ***Hanspeter Pfister*** — *Jan 2024 - May 2024*

**Stanford HCI Group, Stanford Univeristy** — ***California, USA***
*Research intern advised by* ***Jackie Yang*** *and* ***James Landay*** — *Jul 2023 - Sep 2023*

**ZJU International Design Institute, Zhejiang Univeristy** — ***Hangzhou, China***
*Research intern advised by* ***Lingyun Sun*** *and* ***Zejian Li*** — *Mar 2022 - Jun 2023*

## Projects

**WebXR Fundamental 3D Interaction Methods for Object Manipulation** — ***WebXR for 3D interaction***
*[Video] [Github]* — *Spring 2024*
- A WebXR interaction framework for object manipulation methods including translation, rotation and resizing.

**Generative XR Spatial Interfaces for Human Robot Co-exploration** — ***Situated XR Interfaces***
*[Video] [Github]* — *Summer 2025*
- Reconstructing and synthesizing real-world scene via mobile robot on real-time for human space interaction

**Handbot———Apple WWDC2023 Swift Student Challenge** — ***Human-centered GenAI***
*[Video] [Github]* — *Apr 2023*
- A swift playground encouraging critical thinking and editing on LLM and other genAI content in multi-modalities.

**Is raining threes worth it?———NBA 3-pointer Data Visualization** — ***Data Visualization***
*[Video] [Website]* — *Dec 2022*
- An interactive visualization website featuring NBA 3-pointers data and key players, received 17000+ views

## Skills

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|---|---|
| **Programming** | Agentic engineering/WebXR/Unity3D/ROS/Eye Tracking and EEG/AI Model Training |
| **HCI Research** | Quantitative Analysis/Interview Thematic Analysis/Participatory Design and Analysis |
| **Devices** | Smart glasses and XR headsets/Embedded Computers (Raspi and ESP32)/Turtlebot |